AI GRADING FEEDBACK

Graded: 3/10/2026

IBM Developer
SKILLS NETWORK

Winning Space Race with Data Science

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<10 March 2026>

Outline

- Executive Summary
- Introduction
- Methodology
- Results
- Conclusion
- Appendix

Executive Summary

Project Objective:
To predict the success of SpaceX Falcon 9 first-stage landings by analyzing historical launch data, utilizing machine learning to minimize costs associated with rocket reuse.

Summary of Methodologies

- **Data Collection & Wrangling:** Leveraged the SpaceX REST API and web scraping (BeautifulSoup) to compile a comprehensive dataset of 90 records.
- **Exploratory Data Analysis (EDA):** Performed statistical analysis via SQL (IBM Db2) and visualized trends using Seaborn and Matplotlib to identify correlations between payload, orbit, and site.
- **Interactive Analytics:** Developed an interactive map using Folium to analyze launch site geography and a Plotly Dash dashboard for real-time visual filtering.
- **Classification Modeling:** Trained and optimized four classification algorithms (Logistic Regression, SVM, Decision Tree, and KNN) using standardized features and GridSearchCV for hyperparameter tuning.

Summary of Results

- **Geographic Insights:** Launch sites are strategically located on coasts (safety) and at lower latitudes (equatorial boost).
- **Dashboard Discoveries:** Identified specific launch sites with the highest success volumes and isolated the payload ranges most conducive to successful first-stage recovery.
- **Model Performance:** Achieved a consistent predictive accuracy of **83.33%** across multiple models on the test set (18 records).
- **Optimal Algorithm:** Support Vector Machines (SVM) utilizing a **sigmoid kernel** provided the most robust results on the validation dataset.

Introduction

Project Background

•SpaceX reduces launch costs by millions through the innovative reuse of Falcon 9 first-stage boosters.

•While landings are now frequent, mission cost and insurance depend heavily on the ability to predict successful recoveries.

•This project utilizes data science to analyze historical launches and build a robust prediction model for landing outcomes.

Core Problems to Solve

•**Geographical Impact:** How do launch site locations and coastal surroundings influence mission success?

•**Technical Trends:** How has the success rate improved over time with technological maturation?

•**Mission Variables:** To what extent do **Payload Mass** and **Orbit Type** correlate with landing success?

•**Booster Performance:** Which booster versions and landing types (drone ship vs. ground pad) are most reliable?

•**Predictive Capability:** Can machine learning accurately determine landing success using only public mission data?

Section 1

Methodology

Methodology

Executive Summary

- Data collection methodology:
  - Rest API and Web Scrapping
- Perform data wrangling
  - Clean data, feature engineering, null data handling
- Perform exploratory data analysis (EDA) using visualization and SQL
- Perform interactive visual analytics using Folium and Plotly Dash
- Perform predictive analysis using classification models
  - Pre-process, tuning, evaluation through scoring and confusion matrix

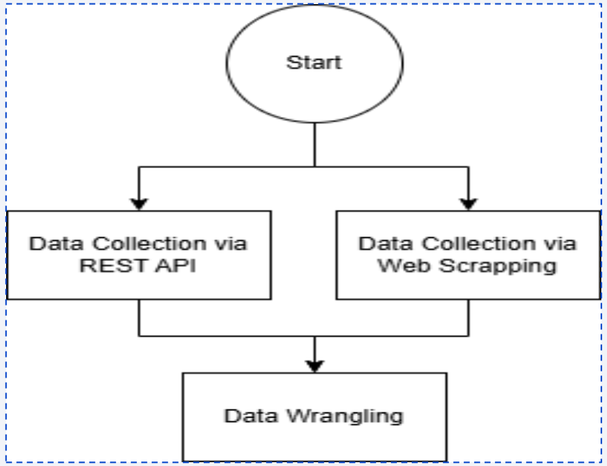
Data Collection

To build the foundation of this analysis, I executed a dual-source data collection strategy to ensure a comprehensive and accurate mission history:

•**REST API Integration:** Utilized the SpaceX API to programmatically retrieve detailed flight records, booster specifications, and landing outcomes for all Falcon 9 missions.

•**Web Scraping:** Employed **BeautifulSoup** to scrape the "List of Falcon 9 and Falcon Heavy launches" from Wikipedia, capturing supplementary technical data for historical missions.

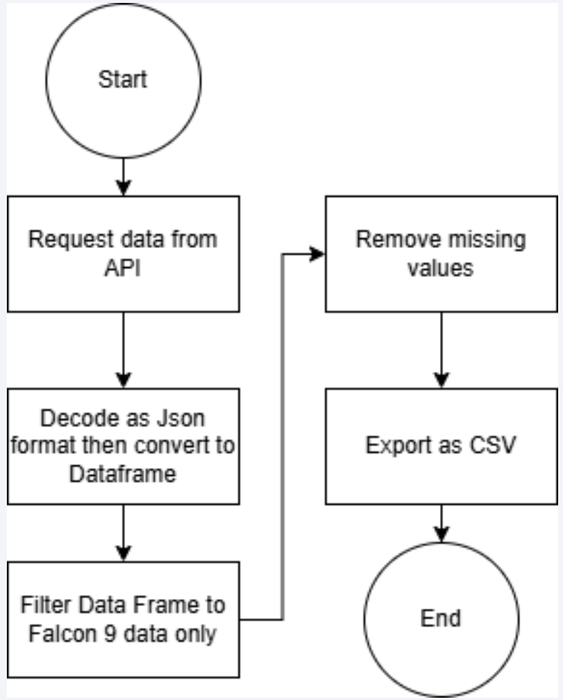
•**Data Consolidation:** Merged both sources into a unified dataset, standardized mission labels (Class), and cleaned inconsistencies to prepare 90 records for modeling.



Data Collection – SpaceX API

- **REST API Integration:** Utilized the SpaceX API to programmatically retrieve detailed flight records, booster specifications, and landing outcomes for all Falcon 9 missions.

- Github url: https://github.com/itslutfi/IBM-final-project/blob/275a05c0a74331d63b94bb6ecd758cae838ef033/01%20jupyter-labs-spacex-data-collection-api.ipynb



Correct Partial Needs Work

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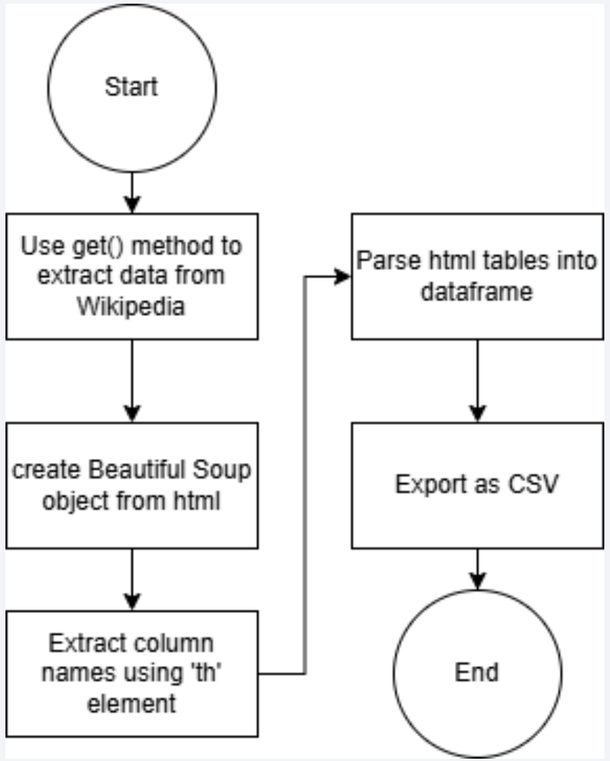
[CORRECT] **Question 1: Did the learner uploaded the URL of GitHub repository, including all the completed notebooks and Python files Question 2: Did the learner upload the final presentation as a .pdf file? Question 3: Did the learner complete the required Executive Summary slide? Question 4: Did the learner complete the required Introduction slide? Question 5: Did the learner complete the required data collection and data wrangling methodology slides? Question 6: Did the learner complete the required EDA and data visualization methodology slides? Question 7: Did the learner complete the predictive analysis (classification) slide? Question 8: Did the learner complete the required EDA with visualization slides? Question 9: Did the learner complete the required EDA with SQL slides/queries? Question 10: Did the learner complete the required Folium map slides? Question 11: Did the learner complete the required Plotly Dash-related slides? Question 12: Did the learner complete the predictive analysis results slides based on the following criteria? Question 13: Did the learner complete the Conclusion slide? The completed Conclusion slide should include at least: Question 14: Did the learner apply any creativity to this presentation? Creativity may include aspects such as: Question 15: Was the learner able to find and display any innovative insights? ** (15/15 points): The presentation includes all required components as per the rubric. The learner uploaded the GitHub URL for the project and all completed notebooks. The presentation includes a well-structured Executive Summary, an Introduction explaining the project context, detailed data collection (both using REST API and web scraping), data wrangling methodology, EDA with visualizations, EDA with SQL, and interactive analytics utilizing Folium and Plotly Dash. Furthermore, predictive analysis has been

Data Collection - Scraping

- **Web Scraping:** Employed **BeautifulSoup** to scrape the "List of Falcon 9 and Falcon Heavy launches" from Wikipedia, capturing supplementary technical data for historical missions
- GitHub URL: https://github.com/itslutfi/IBM-final-project/blob/275a05c0a74331d63b94bb6ecd758cae838ef033/02%20jupyter-labs-webscraping.ipynb

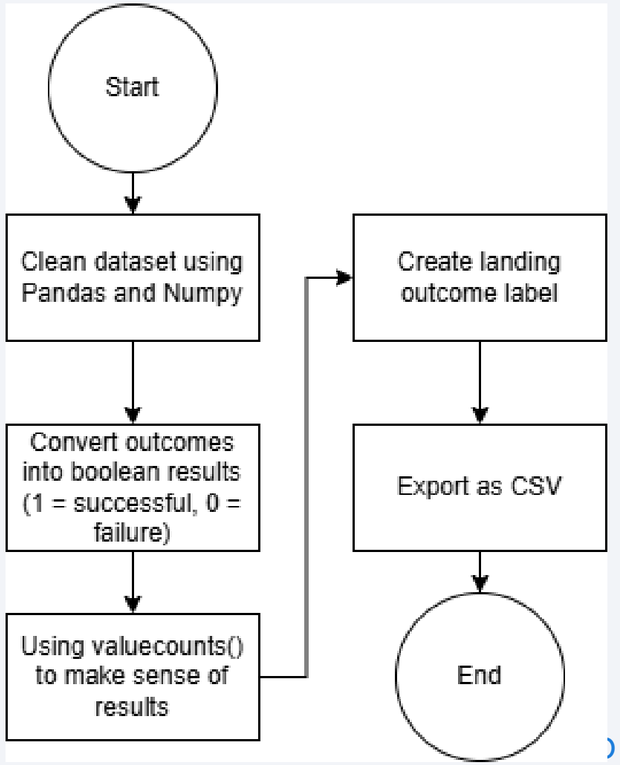


Data Wrangling

Following data collection, the raw records were processed into a structured format suitable for machine learning. The data wrangling phase focused on feature engineering and label creation to define mission success.

- **Outcome Standardization:** Analyzed landing outcomes to create a binary **Class** column, where successful landings (e.g., True ASDS, True RTLS) were assigned **1** and unsuccessful attempts or failures were assigned **0**.
- **Missing Value Management:** Identified null values in the dataset—specifically within landing attempt details—and handled them to ensure a continuous dataset of **90 records**.
- **Feature Identification:** Filtered the dataset to retain critical predictive variables, such as **Payload Mass**, **etc.**

Github: https://github.com/itslutfi/IBM-final-project/blob/275a05c0a74331d63b94bb6ecd758cae838ef033/03%20labs-jupyter-spacex-Data%20wrangling.ipynb



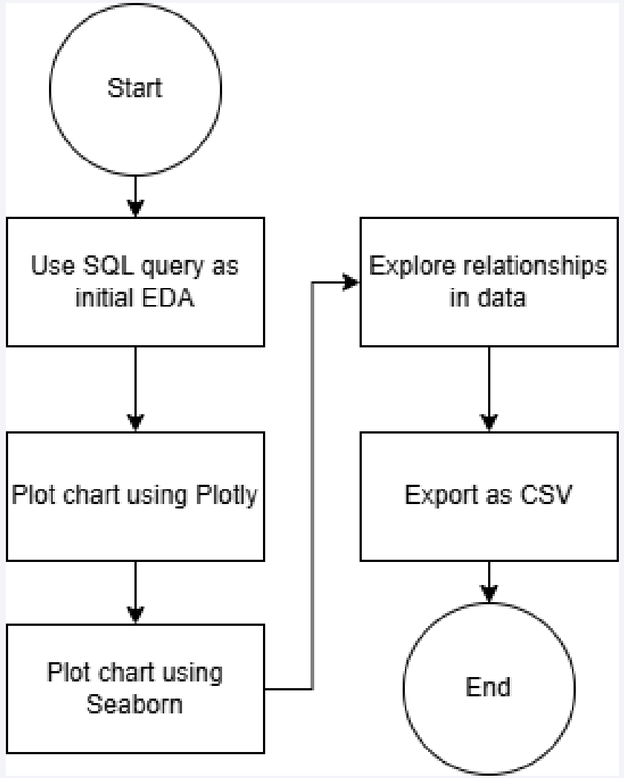
EDA with Data Visualization

To uncover the hidden patterns in SpaceX launch history, I performed **Exploratory Data Analysis (EDA)** using Seaborn and Matplotlib. Each chart was selected to isolate specific variables influencing landing success.

Plotted Charts & Strategic Rationale

- **Scatter Plots (Payload vs. Launch Site):** Used to visualize the distribution of payload masses across different sites and identify if specific locations handle heavier missions with higher success rates.
- **Line Charts (Success Rate vs. Year):** Plotted to track the learning curve of SpaceX technology, demonstrating the steady increase in reliability from 2010 to 2020.
- **Bar Charts (Success Rate vs. Orbit):** Employed to determine which orbital destinations (e.g., LEO, GTO, ISS) are most conducive to first-stage recovery.
- **Catplots (Flight Number vs. Launch Site):** Used to observe the operational frequency and success trends over time at CCFAS, VAFB, and KSC sites.

Github: https://github.com/itslutfi/IBM-final-project/blob/275a05c0a74331d63b94bb6ecd758cae838ef033/04%20edadataviz.ipynb



EDA with SQL

Slide 9: EDA with SQL Results

To perform deeper quantitative analysis, I used **SQL** to query the SpaceX dataset. This allowed for precise filtering and aggregation of launch data to uncover specific operational performance metrics.

SQL Query Summaries

- **Launch Site Identification:** Queried the unique names of the launch sites to establish the geographic baseline for the dataset.
- **Site-Specific Filtering:** Extracted all records for launches originating from sites beginning with "CCA" to analyze Cape Canaveral's specific performance.
- **Payload Mass Calculations:** Calculated the total payload mass carried by NASA-contracted missions to understand the scale of logistical support provided.
- **Averaging Metrics:** Computed the average payload mass for the F9 v1.1 booster version to establish a baseline for technical comparisons.
- **Success Timeline:** Identified the date of the first successful ground pad landing to mark a significant milestone in SpaceX's recovery history.
- **Booster Reliability:** Listed the names of boosters that successfully landed on a drone ship with payload masses between 4,000 and 6,000 kg.
- **Mission Status Counts:** Aggregated total counts for successful and failed mission outcomes to determine the overall program success rate.
- **Max Payload Identification:** Identified the booster versions responsible for carrying the maximum payload mass recorded in the dataset.
- **Landing Outcome Trends:** Selected failed landing records from 2015 to observe the initial challenges in the first-stage recovery process.
- **Site Ranking:** Ranked the launch sites by their total number of successful landings to identify the most productive facilities.

https://github.com/itslutfi/IBM-final-project/blob/ed268a2f97e85da957718afce4ff787dbdc3b0b8/jupyter-labs-eda-sql-coursera_sqllite.ipynb

Build an Interactive Map with Folium

To perform geographical visual analytics, I developed an interactive map using **Folium**. This allowed for a spatial evaluation of SpaceX launch sites relative to their surroundings and coastal proximity.

Interactive Map Objects & Rationale

•**Coordinate Markers:** Plotted specific launch sites (LC-39A, SLC-40, VAFB SLC-4E) to establish the primary operational hubs.

•**Success-Color Circles:** Added labeled circles around sites, color-coded by landing outcome (Green for Success, Red for Failure) to visually identify high-performing locations.

•**Marker Clusters:** Implemented a MarkerCluster to manage overlapping launch markers, ensuring the map remains clean and readable during high-density mission analysis.

•**Distance Poly-Lines:** Drew lines between launch sites and nearby geographical features (coastlines, railways, highways) to calculate the minimum distance to critical infrastructure.

•**Proximity Labels:** Added mouse-over labels showing precise distances in kilometers to demonstrate that sites are strategically placed near water for safety and railways for logistics.

https://github.com/itslutfi/IBM-final-project/blob/6e9a26abd37f1f25f5ce7f0b0b45148c3f03d977/05_lab_jupyter_launch_site_location.ipynb

Build a Dashboard with Plotly Dash

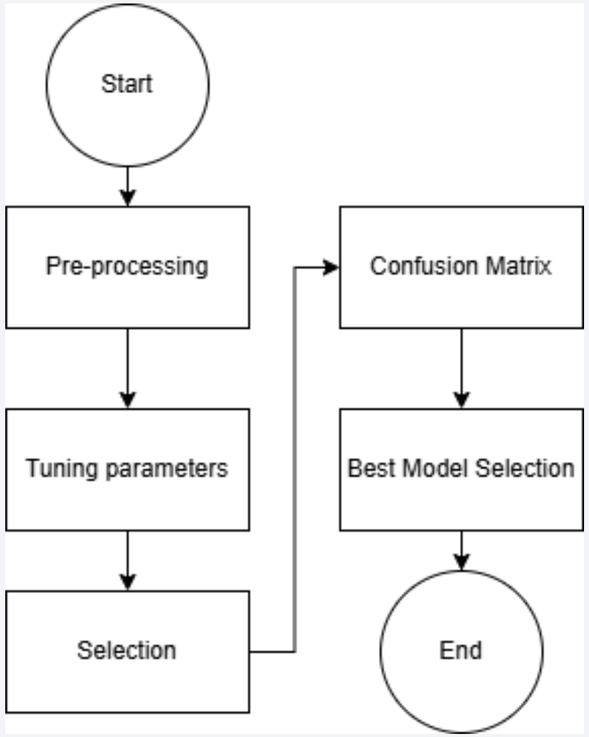
- Summarize what plots/graphs and interactions you have added to a dashboard
- Explain why you added those plots and interactions
- Add the GitHub URL of your completed Plotly Dash lab, as an external reference and peer-review purpose

Predictive Analysis (Classification)

The final phase of the project involved building a machine learning pipeline to predict landing success. I evaluated four distinct classification algorithms to determine which provided the most reliable results for SpaceX's operational data.

Model Development Process

- **Pipeline Foundation:** Standardized the feature matrix X to ensure all variables (e.g., Payload Mass, Flight Number) shared a common scale.
- **Rigorous Evaluation:** Split data into training and test sets (80/20) and utilized **10-fold cross-validation** to ensure model stability.
- **Iterative Improvement:** Performed hyperparameter tuning via **GridSearchCV** to optimize parameters for each algorithm.
- **Performance Benchmarking:** Compared models based on validation accuracy and final test set scores to identify the most robust predictor.



https://github.com/itslutfi/IBM-final-project/blob/ed268a2f97e85da957718afce4ff787dbdc3b0b8/06%20SpaceX_Machine%20Learning%20Prediction_Part_5.ipynb

Results

Exploratory Data Analysis (EDA)

•**Success Trends:** Launch reliability improved significantly after 2013, with 100% success rates in specific orbits like **ES-L1**, **GEO**, and **SSO**.

•**Mission Variables:** Heavier payloads (over 5,000 kg) and specific launch sites like **KSC LC-39A** show distinct correlations with higher success volumes.

Interactive Analytics Demo

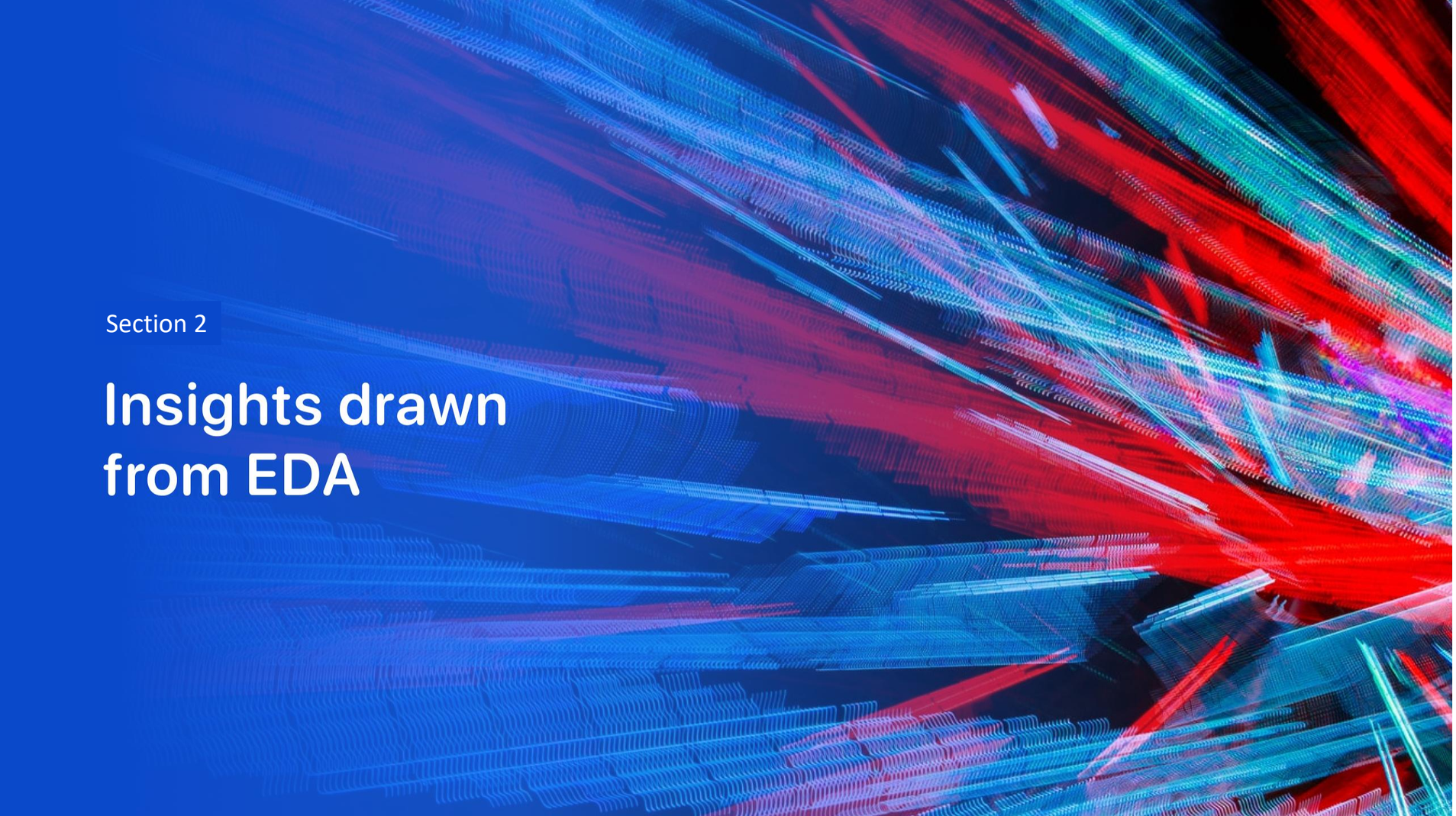
•**Geographic Logic:** Folium maps confirmed all sites are within 5 km of coastlines and railways to optimize safety and logistics.

•**Real-Time Filtering:** The Plotly Dash dashboard allows for dynamic filtering of success rates based on **Payload Range** and **Launch Site**.

Predictive Analysis Results

•**Model Performance:** All classification models achieved a consistent minimum **83.33% accuracy** on the 18-record test set.

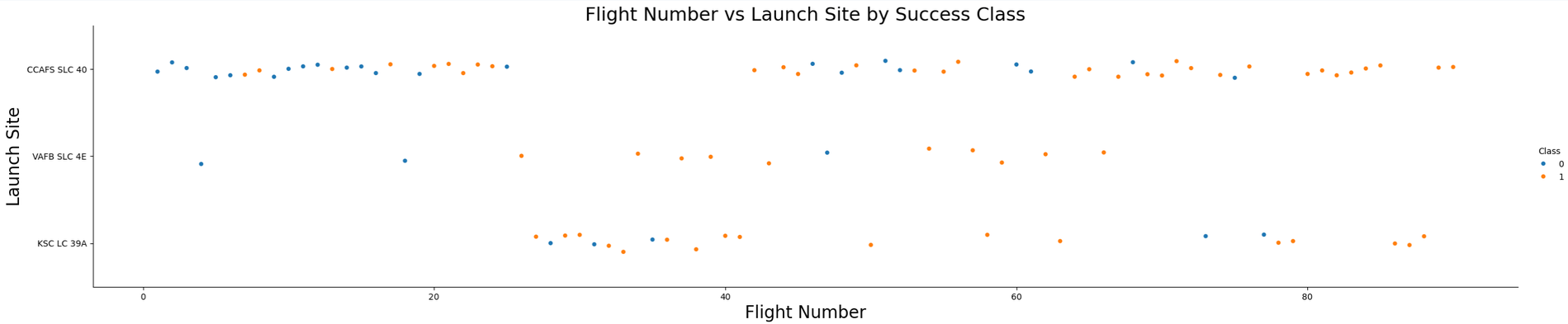
•**Optimal Method: SVM** using a **sigmoid kernel** was identified as the top-performing algorithm during validation.



Section 2

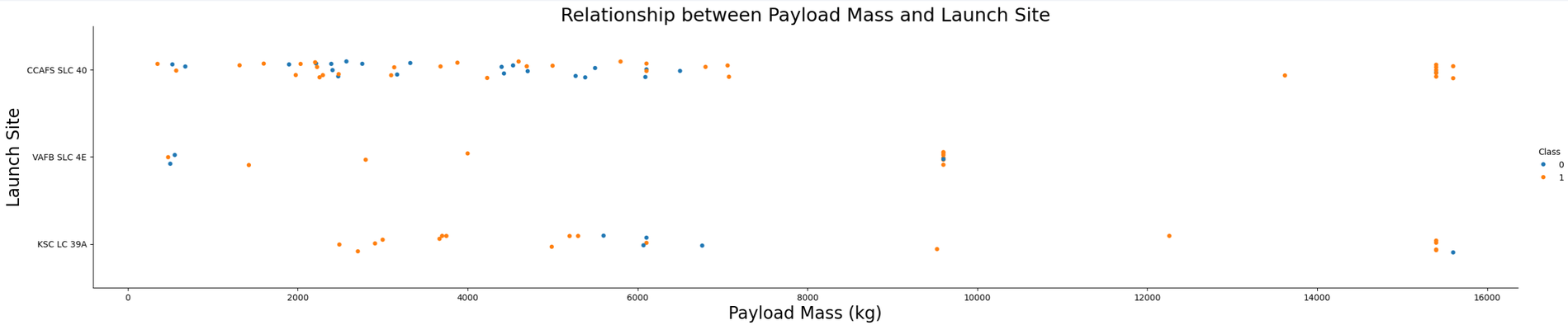
Insights drawn from EDA

Flight Number vs. Launch Site



- **Operational Progression:** The scatter plot (catplot) illustrates that earlier flight numbers were concentrated at **CCAFS SLC 40**, while more recent missions have seen a shift toward **KSC LC 39A** and **VAFB SLC 4E**.
- **Success Density:** As the flight numbers increase, there is a visible increase in the density of successful landings (Class 1) across all sites, confirming the technological maturation identified in the predictive analysis.

Payload vs. Launch Site



- **Site Capacity: VAFB SLC 4E** primarily handles lighter payloads (under 10,000 kg), while **CCAFS SLC 40** and **KSC LC 39A** demonstrate the capacity to launch and land boosters carrying much heavier payloads.
- **Success Correlation:** There is no visible "weight limit" for success; successful landings (Class 1) are consistently achieved across all payload ranges at the Cape Canaveral and Kennedy Space Center sites.

Correct Partial Needs Work

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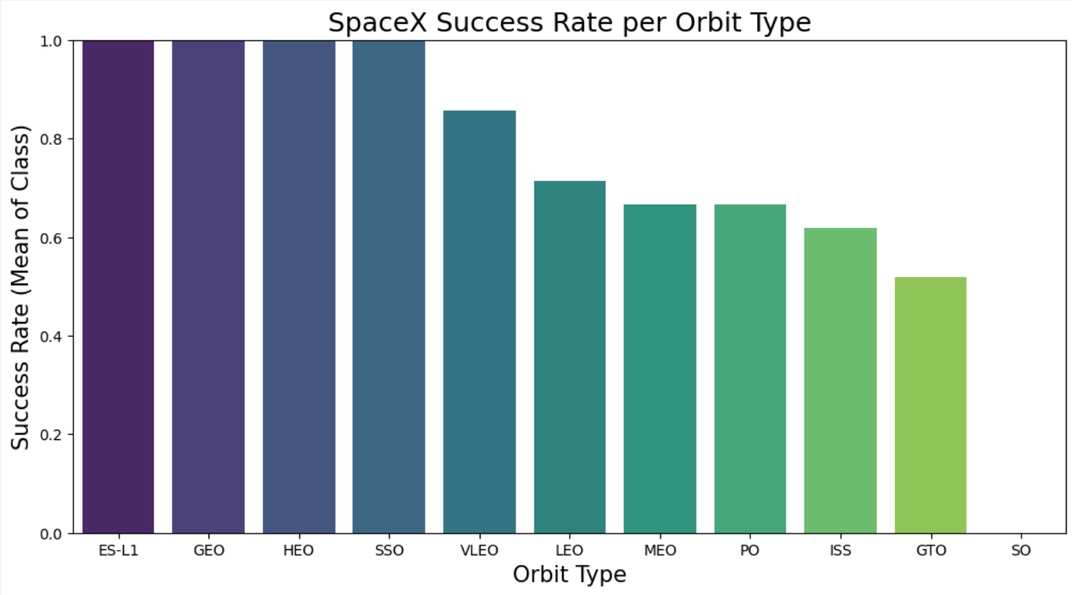
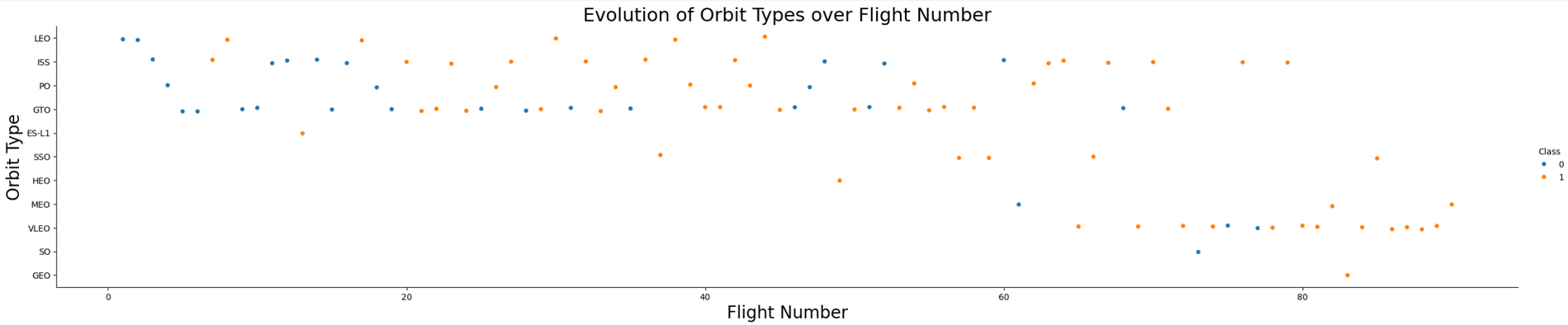
[CORRECT] **Question 1: Did the learner uploaded the URL of GitHub repository, including all the completed notebooks and Python files Question 2: Did the learner upload the final presentation as a .pdf file? Question 3: Did the learner complete the required Executive Summary slide? Question 4: Did the learner complete the required Introduction slide? Question 5: Did the learner complete the required data collection and data wrangling methodology slides? Question 6: Did the learner complete the required EDA and data visualization methodology slides? Question 7: Did the learner complete the predictive analysis (classification) slide? Question 8: Did the learner complete the required EDA with visualization slides? Question 9: Did the learner complete the required EDA with SQL slides/queries? Question 10: Did the learner complete the required Folium map slides? Question 11: Did the learner complete the required Plotly Dash-related slides? Question 12: Did the learner complete the predictive analysis results slides based on the following criteria? Question 13: Did the learner complete the Conclusion slide? The completed Conclusion slide should include at least: Question 14: Did the learner apply any creativity to this presentation? Creativity may include aspects such as: Question 15: Was the learner able to find and display any innovative insights? ** (15/15 points): The presentation includes all required components as per the rubric. The learner uploaded the GitHub URL for the project and all completed notebooks. The presentation includes a well-structured Executive Summary, an Introduction explaining the project context, detailed data collection (both using REST API and web scraping), data wrangling methodology, EDA with visualizations, EDA with SQL, and interactive analytics utilizing Folium and Plotly Dash. Furthermore, predictive analysis has been

Success Rate vs. Orbit Type

- **High-Performance Orbits:** Missions to **ES-L1, GEO, HEO,** and **SSO** achieved a perfect 100% success rate, indicating these mission profiles are highly optimized for first-stage recovery.
- **Lower Success Profiles:** The **GTO** orbit shows a success rate of approximately 50%, highlighting the increased technical difficulty and fuel constraints associated with Geostationary Transfer Orbits.

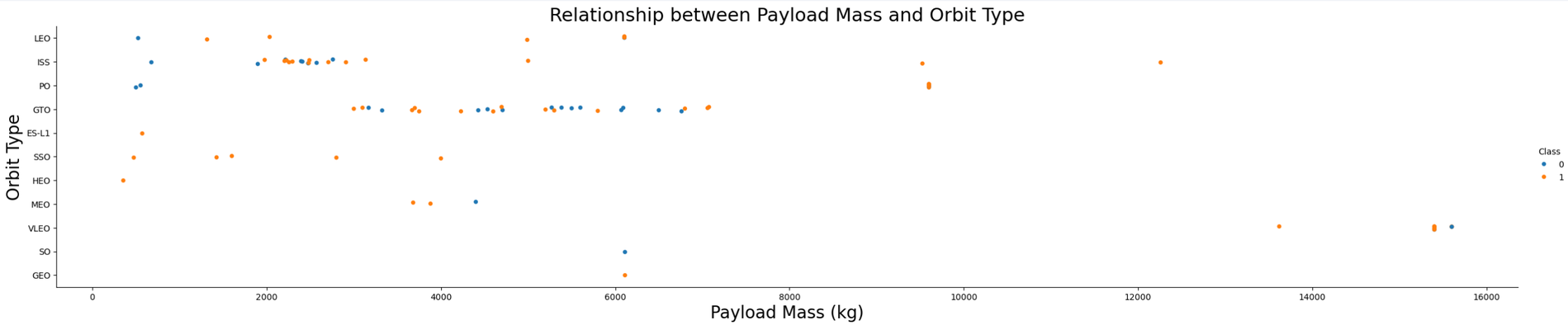


Flight Number vs. Orbit Type



•**Success Progression:** The plot reveals that as the flight number increases, the frequency of successful landings (Class 1) improves across various orbits, particularly in high-demand paths like **GTO** and **ISS**.
•**Orbit Specialization:** Early missions were concentrated in **LEO** and **ISS** orbits, while more recent, higher flight numbers show a transition into more diverse and complex orbital profiles like **SSO** and **Polar**

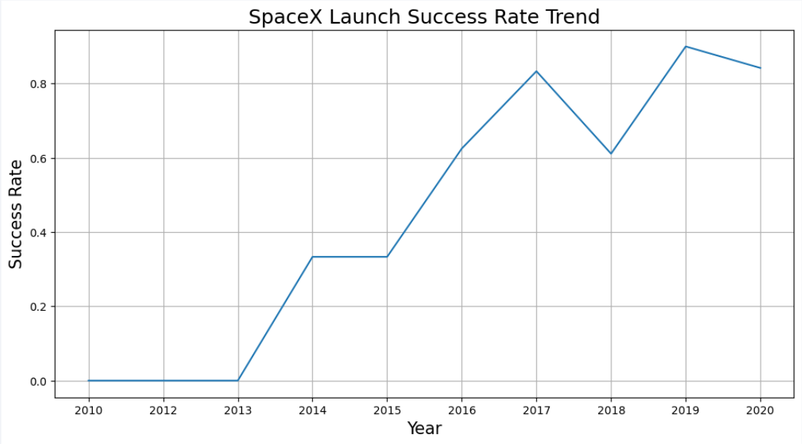
Payload vs. Orbit Type



- **Payload Clustering:** Heavy payloads are predominantly focused on **GTO** and **PO** (Polar) orbits, while lighter payloads are more common for **ISS** and **LEO** missions.
- **Recovery Success:** The visualization highlights that for **GTO** orbits, success is achieved even with significantly higher payload masses, whereas smaller payloads in **LEO** and **ISS** orbits show very high reliability.

Launch Success Yearly Trend

- **Learning Curve Visualization:** The line chart reveals a clear upward trend in landing success, starting at 0% in 2010 and reaching a sustained peak near 90–100% by 2020.
- **Technological Maturity:** The sharp increase between 2013 and 2017 highlights the period where SpaceX successfully transitioned from experimental recovery attempts to consistent operational landings.



All Launch Site Names

SQL Query Explanation

- **The Query:** SELECT DISTINCT Launch_Site FROM SPACEXTBL;
- **Key Results:** The query identifies four distinct site labels used in the database. It highlights that operations are primarily split between the **Florida coasts** (CCAFS/KSC) and the **California coast** (VAFB), providing the geographic diversity needed for various mission orbits.

Launch_Site
CCAFS LC-40
VAFB SLC-4E
KSC LC-39A
CCAFS SLC-40

Launch Site Names Begin with 'CCA'

SQL Query Explanation

•**The Query:** SELECT * FROM SPACEXTBL WHERE Launch_Site LIKE 'CCA%' LIMIT 5;.

•**Key Results:** These records represent the "Dragon" era of SpaceX. The results show that early payloads were significantly lighter (under 1,000 kg) compared to modern missions and primarily focused on **Low Earth Orbit (LEO)** for NASA resupply objectives.

Date	Time (UTC)	Booster_Version	Launch_Site	Payload
2010-06-04	18:45:00	F9 v1.0 B0003	CCAFS LC-40	Dragon Spacecraft Qualification Unit
2010-12-08	15:43:00	F9 v1.0 B0004	CCAFS LC-40	Dragon demo flight C1, two CubeSats, barrel of Brouere cheese
2012-05-22	7:44:00	F9 v1.0 B0005	CCAFS LC-40	Dragon demo flight C2
2012-10-08	0:35:00	F9 v1.0 B0006	CCAFS LC-40	SpaceX CRS-1
2013-03-01	15:10:00	F9 v1.0 B0007	CCAFS LC-40	SpaceX CRS-2

Total Payload Mass

SUM(PAYLOAD_MASS__KG_)
45596

SQL Query Explanation

•**The Query:** SELECT SUM(PAYLOAD_MASS__KG_) FROM SPACEXTBL WHERE Customer LIKE '%NASA%';.

•**Key Insight:** The result shows that NASA-contracted missions account for over **45 metric tons** of equipment, supplies, and scientific hardware delivered to orbit. This substantial figure highlights the critical role SpaceX plays in supporting ISS resupply missions and international space research

Average Payload Mass by F9 v1.1

AVG(PAYLOAD_MASS__KG_)
2534.6666666666665

SQL Query Explanation

•**The Query:** SELECT AVG(PAYLOAD_MASS__KG_) FROM SPACEXTBL WHERE Booster_Version = 'F9 v1.1';

•**Key Insight:** This result provides a technical baseline for the "v1.1" generation. The average mass of ~2,535 kg demonstrates the booster's role during the transition period of the Falcon 9 program, primarily handling medium-class payloads before the introduction of the "Full Thrust" variants.

First Successful Ground Landing Date

MIN(Date)
2015-12-22

SQL Query Explanation

•**The Query:** SELECT MIN(Date) FROM SPACEXTBL WHERE Landing_Outcome = 'Success (ground pad)';

•**Key Insight:** This result marks the historic first return of a Falcon 9 orbital-class booster to a landing zone on solid ground. Occurring at Cape Canaveral, this success proved the commercial and technical feasibility of rapid rocket reuse, a core objective of the SpaceX program.

Successful Drone Ship Landing with Payload between 4000 and 6000

Booster_Version	PAYLOAD_MASS__KG_	Landing_Outcome
F9 FT B1022	4696	Success (drone ship)
F9 FT B1026	4600	Success (drone ship)
F9 FT B1021.2	5300	Success (drone ship)
F9 FT B1031.2	5200	Success (drone ship)

- **The Query:** SELECT Booster_Version, PAYLOAD_MASS__KG_, Landing_Outcome FROM SPACEXTBL WHERE Landing_Outcome = 'Success (drone ship)' AND PAYLOAD_MASS__KG_ BETWEEN 4001 AND 5999;
- **Key Insight:** These four records highlight the capability of the **Falcon 9 Full Thrust (FT)** variant to successfully recover the first stage even when carrying substantial payloads to high-energy orbits. This specific weight range often corresponds to Geostationary Transfer Orbits (GTO), where drone ship landings are technically required due to fuel constraints.

Total Number of Successful and Failure Mission Outcomes

Mission_Outcome	Total_Count
Failure (in flight)	1
Success	98
Success	1
Success (payload status unclear)	1

SQL Query Explanation

•**The Query:** SELECT Mission_Outcome, COUNT(*) AS Total_Count FROM SPACEXTBL GROUP BY Mission_Outcome;

•**Key Insight:** The results demonstrate an exceptional success rate for the SpaceX Falcon 9 program. With nearly 100 successful missions compared to only one recorded in-flight failure, the data validates the program's transition from an experimental phase to a highly reliable orbital launch service. The "Success" variations are typically due to minor data entry differences or specific payload nuances, but they all represent successful orbital deliveries.

Boosters Carried Maximum Payload

SQL Query Explanation

•**The Query:** SELECT Booster_Version, PAYLOAD_MASS__KG_ FROM SPACEXTBL WHERE PAYLOAD_MASS__KG_ = (SELECT MAX(PAYLOAD_MASS__KG_) FROM SPACEXTBL);

•**Key Insight:** The result reveals that the "Block 5" (B5) version is the primary workhorse for SpaceX's heaviest missions. Specifically, these 15.6-metric-ton payloads frequently correspond to **Starlink** satellite deployments, where maximizing the number of satellites per launch is critical for constellation build-out efficiency.

Booster_Version	PAYLOAD_MASS__KG_
F9 B5 B1048.4	15600
F9 B5 B1049.4	15600
F9 B5 B1051.3	15600
F9 B5 B1056.4	15600
F9 B5 B1048.5	15600
F9 B5 B1051.4	15600
F9 B5 B1049.5	15600
F9 B5 B1060.2	15600
F9 B5 B1058.3	15600
F9 B5 B1051.6	15600
F9 B5 B1060.3	15600
F9 B5 B1049.7	15600

2015 Launch Records

Month_Name	Landing_Outcome	Booster_Version	Launch_Site
January	Failure (drone ship)	F9 v1.1 B1012	CCAFS LC-40
April	Failure (drone ship)	F9 v1.1 B1015	CCAFS LC-40

SQL Query Explanation

•**The Query:** SELECT MonthName(Date), Landing_Outcome, Booster_Version, Launch_Site FROM SPACEXTBL WHERE Landing_Outcome = 'Failure (drone ship)' AND YEAR(Date) = 2015;

•**Key Insight:** These two records represent the pioneering "hit-and-miss" phase of autonomous sea landings. Both attempts originated from Cape Canaveral using the **F9 v1.1** booster. While these missions resulted in "hard landings" on the drone ship, the data gathered from these failures was instrumental in refining the navigation and throttle control algorithms that led to the first successful sea landing in 2016.

Rank Landing Outcomes Between 2010-06-04 and 2017-03-20

SQL Query Explanation

•**The Query:** SELECT Landing_Outcome, COUNT(*) AS Outcome_Count FROM SPACEXTBL WHERE Date BETWEEN '2010-06-04' AND '2017-03-20' GROUP BY Landing_Outcome ORDER BY Outcome_Count DESC;

•**Key Insight:** During this period, "No attempt" was the most common outcome, reflecting the initial focus on primary mission success over booster recovery. However, the data also captures the balanced "trial and error" phase of drone ship landings, with an equal number of successes and failures (5 each), demonstrating the rapid iterative testing that eventually led to the high reliability seen in modern SpaceX operations.

Landing_Outcome	Outcome_Count
No attempt	10
Success (drone ship)	5
Failure (drone ship)	5
Success (ground pad)	3
Controlled (ocean)	3
Uncontrolled (ocean)	2
Failure (parachute)	2
Precluded (drone ship)	1

Section 3

Launch Sites Proximities Analysis

Launch Site Location

- **Site Clustering:** The map reveals a strategic split between the **East Coast (Cape Canaveral and Kennedy Space Center)** and the **West Coast (Vandenberg AFB)**. This allows SpaceX to cover a wide range of orbital inclinations, from standard equatorial orbits to polar and sun-synchronous orbits.

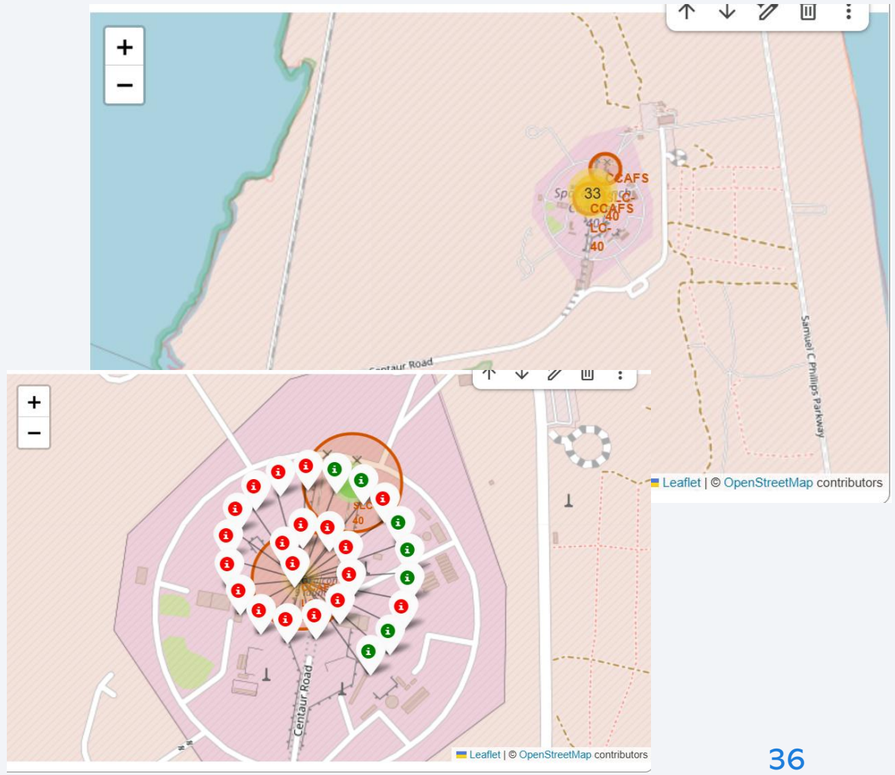
- **Coastal Proximity:** All identified markers are located immediately adjacent to the ocean. This is critical for two reasons: it provides a clear, unpopulated "debris corridor" for ascending rockets and enables the recovery of first-stage boosters on **Autonomous Spaceport Drone Ships (ASDS)** positioned downrange in the Atlantic or Pacific.

Visualizing Launch Success by Site Location

- **Color-Coded Outcomes:** The map utilizes a MarkerCluster where individual landing attempts are color-labeled: **Green markers** represent successful landings, while **Red markers** indicate failed attempts. This allows for an immediate visual assessment of site-specific reliability.

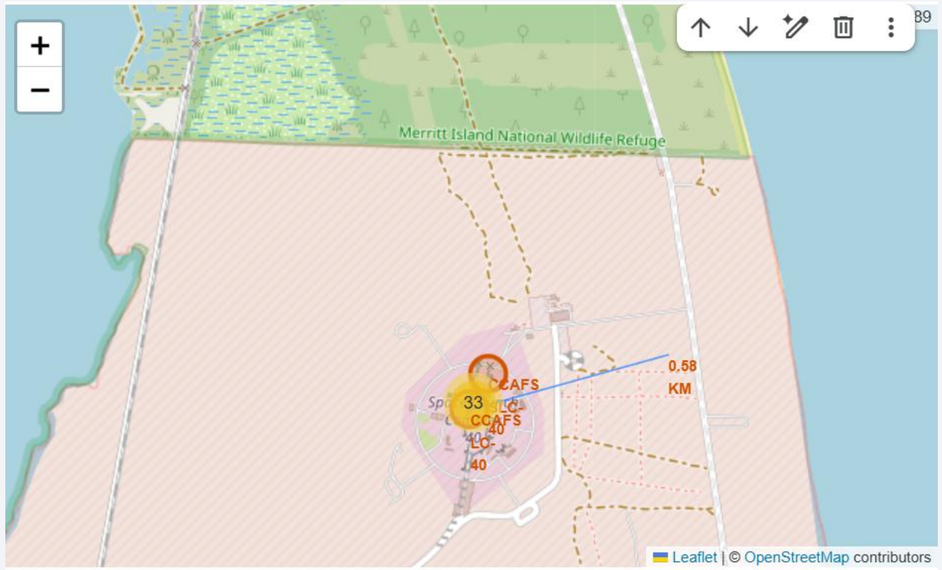
- **Success Density:** By observing the clusters, it is evident that **KSC LC-39A** and **CCAFS SLC-40** have a high density of green markers, reflecting their role in the majority of successful Falcon 9 recoveries.

- **Spatial Patterns:** The visualization confirms that all attempts—regardless of outcome—are aligned with the coastline, ensuring that failed landings occur over open water rather than populated areas, adhering to strict safety protocols.



Proximity Analysis - Distances to Critical Infrastructure

•**Distance Poly-Lines:** The map features drawn lines connecting the **KSC LC-39A** launch site to nearby geographical landmarks, such as the coastline, highways, and the city of Titusville.
•**Safety Buffers:** Findings show that launch sites are intentionally situated at significant distances from heavy population centers (e.g., >15 km from major cities) to minimize noise impact and risk during an anomaly.
•**Logistical Efficiency:** Despite the safety buffer, sites are located within **1–5 km of railways and highways**, facilitating the transport of massive rocket stages and specialized fuel components.
•**Coastal Advantage:** The calculated distance to the coastline is minimal (often <1 km), ensuring that the primary flight path is immediately over the ocean, which is essential for both public safety and downrange recovery operations.

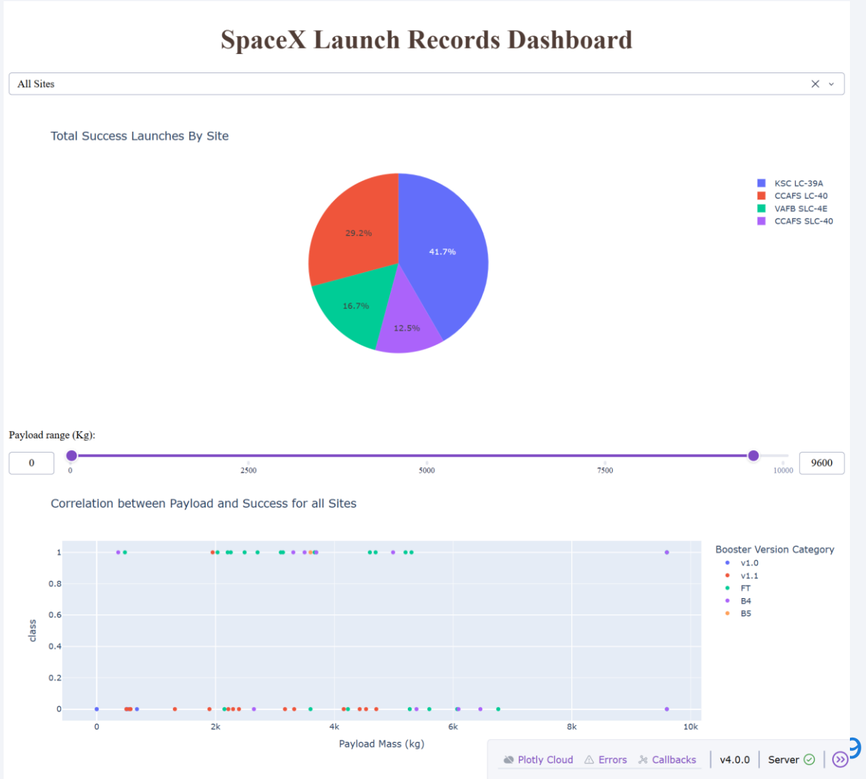


Section 4

Build a Dashboard with Plotly Dash

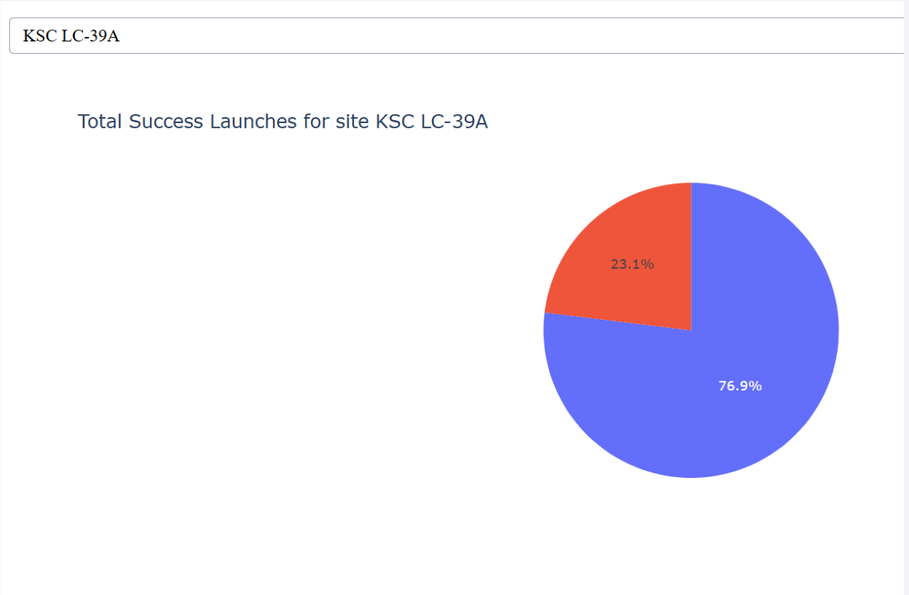
SpaceX Interactive Dashboard - Total Launch Success by Site

- **Top Performing Site**: According to the pie chart, **KSC LC-39A** accounts for the largest proportion of successful launches at **41.7%**.
- **Geographic Distribution**: **CCAFS LC-40** follows with **29.2%** of total successes, while **VAFB SLC-4E** and **CCAFS SLC-40** contribute **16.7%** and **12.5%** respectively.
- **Booster Reliability**: The scatter plot indicates that the **B5** booster version maintains a high success rate across a wide range of payload masses, appearing consistently at the top of the y-axis (class=1).
- **Payload Correlation**: While successes occur across all mass ranges, there is a visible clustering of successful missions for payloads between **2,000 kg and 4,000 kg**.

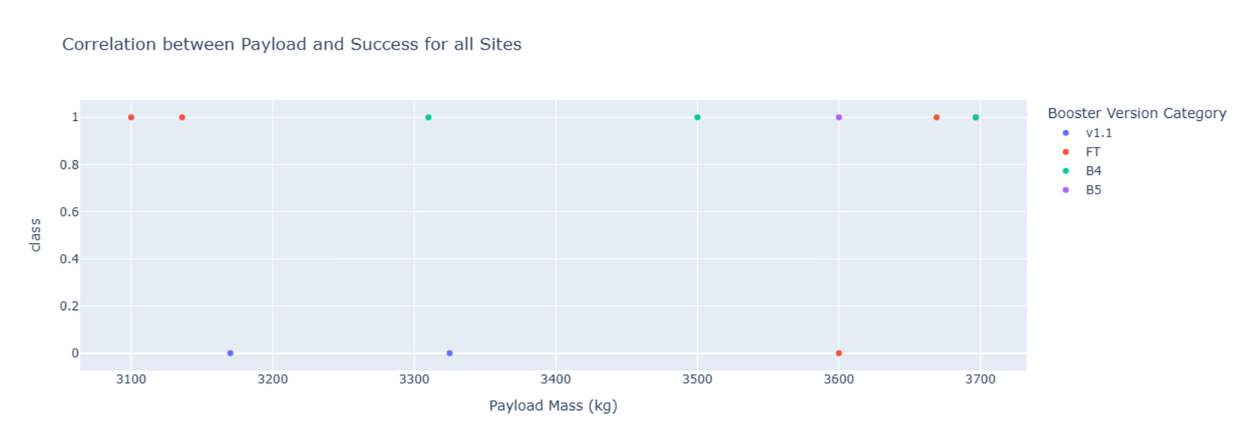


SpaceX Interactive Dashboard - Performance for Highest Success Site

- **Highest Success Rate:** KSC LC-39A maintains a highly impressive success rate of **76.9%**. Only **23.1%** of missions from this site resulted in a landing failure.
- **Payload Reliability:** Successful missions (class 1) are well-distributed across the **2,500 kg to 5,500 kg** payload range.
- **Failure Threshold:** The scatter plot reveals a specific cluster of failures (class 0) for heavy payloads exceeding **5,500 kg** at this site.
- **Booster Dominance:** The **FT (Full Thrust)** and **B5 (Block 5)** versions demonstrate nearly perfect success records at this location, consistently appearing at the y=1 (Success) line for various payload weights.



SpaceX Interactive Dashboard - Correlation Between Payload and Success



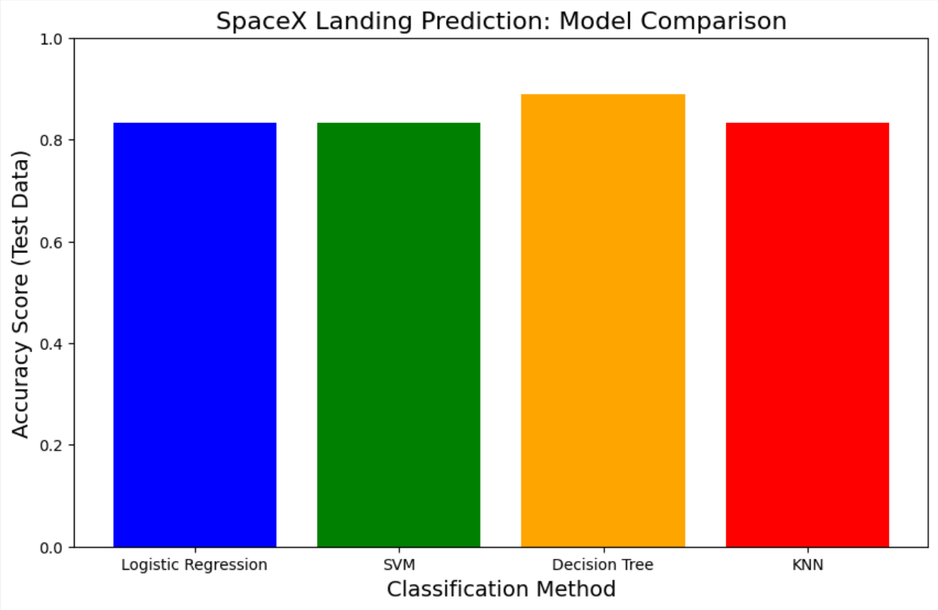
- **Optimum Reliability:** This specific payload range appears to be the most successful for the Falcon 9 fleet, with a high density of markers clustered at the **y=1 (Success)** level.
- **Booster Success Rates:** * **FT (Full Thrust):** Demonstrates high reliability, though it records one specific failure at exactly 3,600 kg.
- **B4 & B5 Versions:** These newer-generation boosters achieve a **100% success rate** within this payload window, indicating significant improvements in landing technology.
- **Mass Correlation:** Successes are consistently found between **3,100 kg and 3,700 kg**, suggesting that the current landing algorithms and fuel reserves are highly optimized for this specific mass profile.

Section 5

Predictive Analysis (Classification)

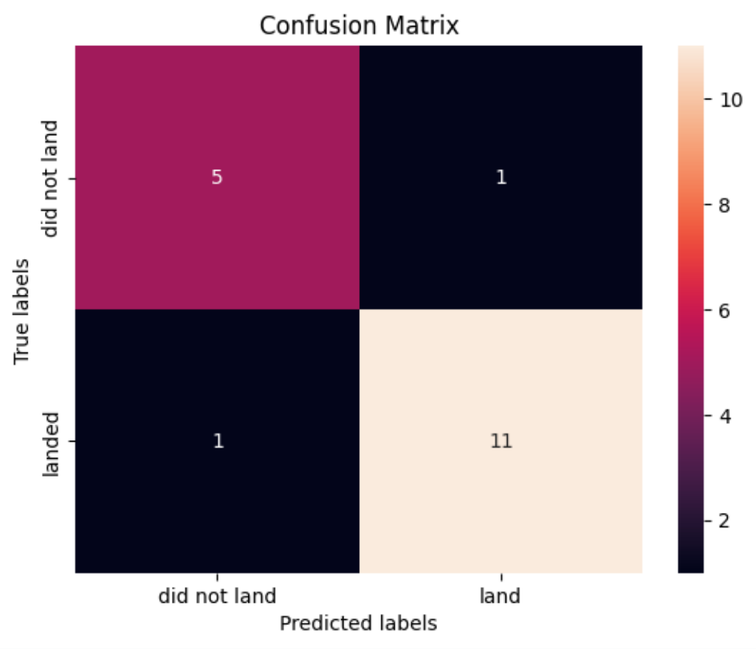
Classification Accuracy

- **Best Performing Method:** The **Decision Tree** algorithm is the optimal choice for this dataset, as it provides the highest predictive power for mission success.
- **Model Robustness:** While the other three models (LR, SVM, KNN) showed identical performance on the test set, the Decision Tree's higher score suggests it more effectively modeled the specific features of the SpaceX launch environment.
- **Strategic Utility:** Utilizing this model allows stakeholders to calculate the financial risk of a mission by predicting the likelihood of booster recovery with nearly **89%** accuracy.



Confusion Matrix

•**High Precision and Recall:** With 16 out of 18 total test cases predicted correctly, the model demonstrates strong reliability for both success and failure scenarios.
•**Balanced Error Rate:** The model shows no specific bias toward over-predicting success or failure, as both the false positive and false negative counts are identical at 1.
•**Impact on Decision Making:** For SpaceX, this matrix validates that the Decision Tree can be used as a high-confidence tool for mission risk assessment, correctly identifying successful landings approximately **89%** of the time.



Conclusions

•**Model Superiority and Reliability:** The **Decision Tree** algorithm is the most effective predictive tool for mission success, achieving a peak test accuracy of **0.8888888888888888**. The confusion matrix confirms its high precision, with only two misclassifications across the entire test set.

•**Optimal Operational Conditions:** Visual analytics reveal an "optimum range" for booster recovery between **3,000 kg and 4,000 kg**. Within this segment, modern booster versions like **B5** maintain a perfect success record, whereas older versions like **v1.1** consistently fail to achieve successful landings.

•**Geographic Performance Leaders: KSC LC-39A** stands as the premier launch site, contributing **41.7%** of all successful SpaceX launches globally. Its individual success rate of **76.9%** underscores its strategic importance for high-value missions.

•**Technological Evolution:** The transition from early Falcon 9 versions to the **Block 5 (B5)** architecture has successfully minimized the correlation between heavy payloads and mission failure. This maturity allows SpaceX to offer highly reliable, reusable launch services that significantly reduce the cost of access to space.

Appendix

- Include any relevant assets like Python code snippets, SQL queries, charts, Notebook outputs, or data sets that you may have created during this project

Thank you!

GRADING SUMMARY

Overall Correctness Score: 100%

Feedback Breakdown:

[+] Correct: 4

[~] Partially Correct: 0

[X] Needs Improvement: 0

Detailed Feedback Summary:

How to Use This Feedback:

* Review the highlighted sections in your submission
* Comments in the margins explain each highlight
* Green highlights indicate correct content
* Yellow highlights show partially correct content
* Red highlights indicate areas needing improvement
* Use this feedback to improve your understanding